


# NPTEL Course on
# "Computer System Optimization for ML"

Lecture #45

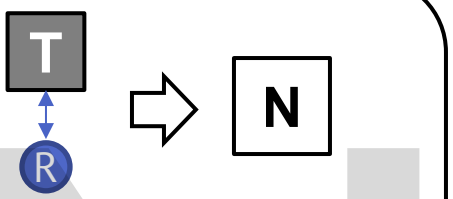
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# Agenda

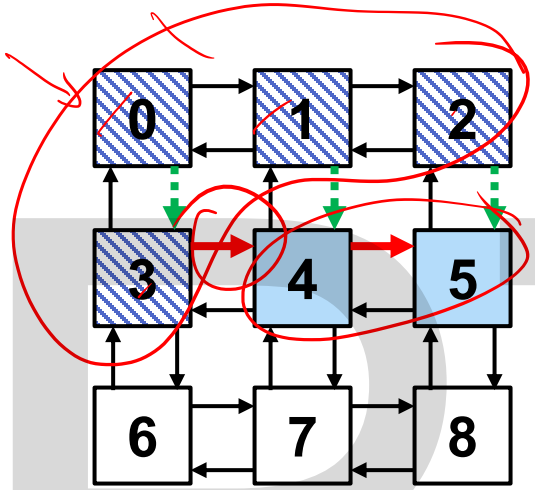
- Recap
- Achieving minimum communication latency

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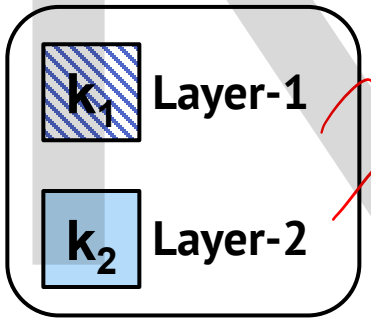
# Congestion-free Communication for DNN




**One node denotes combination of one IMC tile and a router**




- Colored arrows show the link used for the communication
- Dashed (green) arrows show the links with no congestion
- Solid (red) arrows show the links with congestions
  - Example: The link between Node 3 and Node 4 is used by 0-4 and 3-4 traffic




Assumptions

- Y-X routing for NoC
- Each tiles generates P packets to each other tile
- Layer-1 is mapped into Node 0-3; Layer-2 is mapped into Node 4-5
- Communication between Layer-1 and Layer-2

| **Schedule** | **Src→Dest** |
|---|---|
| Cycle-1--Cycle-P | 0→3 |
| Cycle-1--Cycle-P | 3→4 |
| Cycle-1--Cycle-P | 1→5 |
| Cycle-1--Cycle-P | 2→5 |
| (P + 1)--(2P) | 0→4 |
| (P + 1)--(2P) | 4→5 |

Should not 'overlap'!!

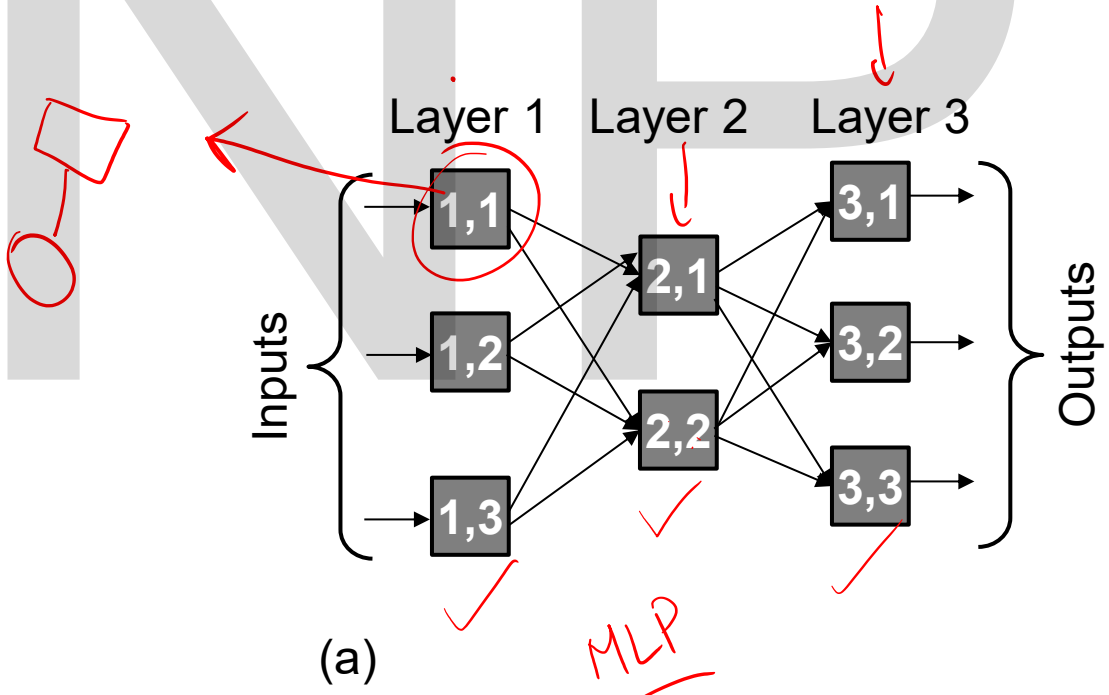
# Agenda

- Recap
- **Achieving minimum communication latency**

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# Achieving Minimum Possible Communication Latency: Example

- **Target: to achieve minimum possible communication latency for a given DNN on IMC-based hardware platform**
- **Example of *minimum possible communication latency***



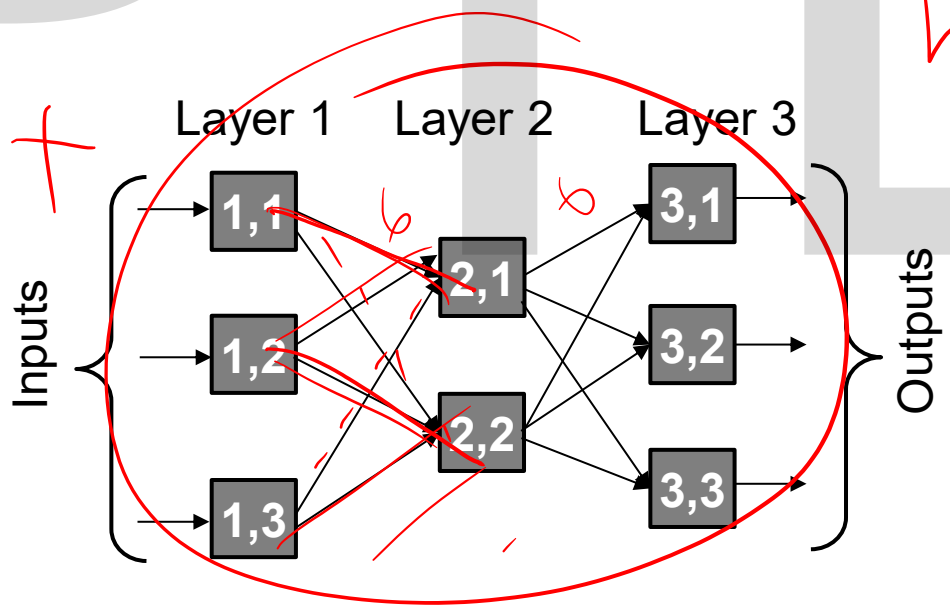

(a)

| Layer 1 – Layer 2 | | Layer 2 – Layer 3 | |
|---|---|---|---|
| Link | Cycle | Link | Cycle |
| $L_{1,1} - L_{2,1}$ | 1 | $L_{2,1} - L_{3,1}$ | 4 |
| $L_{1,2} - L_{2,2}$ | 1 | $L_{2,2} - L_{3,2}$ | 4 |
| $L_{1,1} - L_{2,2}$ | 2 | $L_{2,1} - L_{3,2}$ | 5 |
| $L_{1,3} - L_{2,1}$ | 2 | $L_{2,2} - L_{3,1}$ | 5 |
| $L_{1,2} - L_{2,1}$ | 3 | $L_{2,2} - L_{3,2}$ | 6 |
| $L_{1,3} - L_{2,2}$ | 3 | $L_{2,1} - L_{3,3}$ | 6 |

(b)

# Key Insights

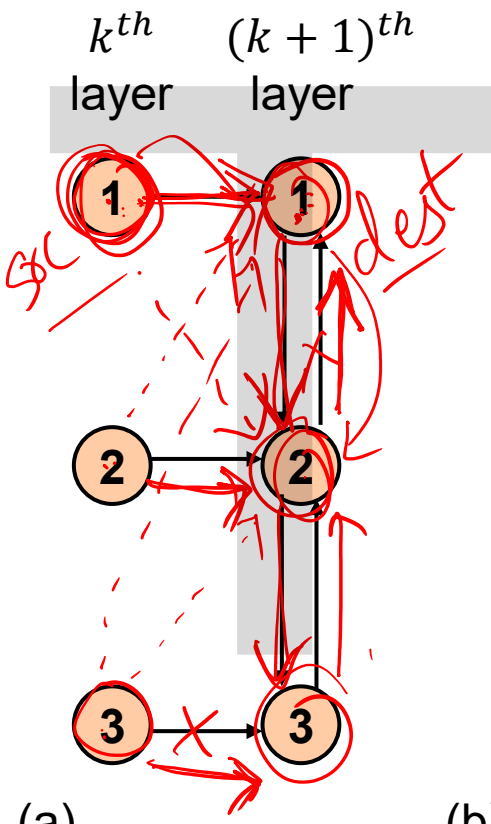
- A single router can not send/receive more than one packet in a cycle. However, multiple routers can send/receive packets in parallel.
- Since a router can send/receive only one packet in a cycle, the congestion is minimum i.e., no more than a transaction is scheduled through a particular link.




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# Solution: Custom Interconnect (A Simple Case)

- Two consecutive layers consists of equal number of routers
  - Three routers in this case
- Each router can send/receive only one packet at one cycle
- In one round of communication each routers in $k^{th}$ layer send a packet to each router in $(k+1)^{th}$ layer
- One transaction is finished in 3 cycles
  - Minimum possible latency




(a)

(b)

| Link | Cycle | Transaction |
|---|---|---|
| $L_{k,1} - L_{(k+1),1}$ | 1 | $R_{k,1} - R_{(k+1),1}$ |
| $L_{k,2} - L_{(k+1),2}$ | 1 | $R_{k,2} - R_{(k+1),2}$ |
| $L_{k,3} - L_{(k+1),3}$ | 1 | $R_{k,3} - R_{(k+1),3}$ |
| $L_{(k+1),1} - L_{(k+1),2}$ | 2 | $R_{k,1} - R_{(k+1),2}$ |
| $L_{(k+1),2} - L_{(k+1),3}$ | 2 | $R_{k,2} - R_{(k+1),3}$ |
| $L_{(k+1),3} - L_{(k+1),2}$ | 2 | $R_{k,3} - R_{(k+1),2}$ |
| $L_{(k+1),2} - L_{(k+1),1}$ | 2 | $R_{k,2} - R_{(k+1),1}$ |
| $L_{(k+1),2} - L_{(k+1),3}$ | 3 | $R_{k,1} - R_{(k+1),3}$ |
| $L_{(k+1),2} - L_{(k+1),1}$ | 3 | $R_{k,3} - R_{(k+1),1}$ |

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# Proof by Induction

**New Transactions:**

**Horizontal Link** ($L_{1N}$-$L_{2N}$) will be active on cycle 1 which does not overlap with any of the transaction of the configuration with (N-1) routers

**Upward Vertical Link:** New transactions will carry the packet from the router at the position (1,N). The vertical link ($L_{2,N}$-$L_{2,(N-1)}$) will be activated on cycle 1. All other new transactions will happen 1 cycle after the last transaction for the configuration with (N-1) routers.

**Downward Vertical Link:** New transactions will happen only in ($L_{2,(N-1)} - L_{2,N}$)

**Schedules in general:**

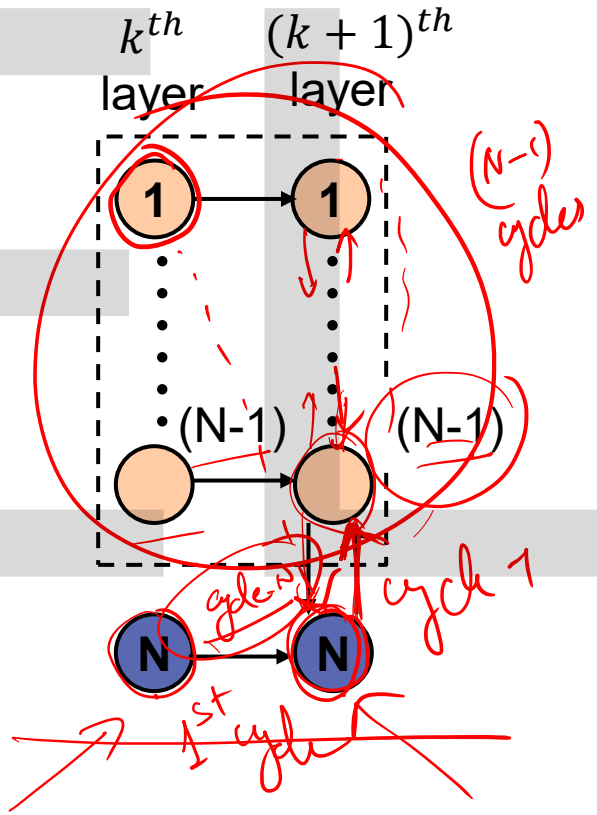
**Horizontal Link:** $L_{k,n}$-$L_{k+1,n}$ in cycle 1 carries transaction $T_{k,n}$-$T_{k+1,n}$

**Upward Vertical Link:** $L_{k+1,n}$-$L_{k+1,n-1}$ in $m^{th}$ cycle carries the transaction

$T_{k,n+m-2}$-$T_{k+1,n-1}$, if the transaction exists

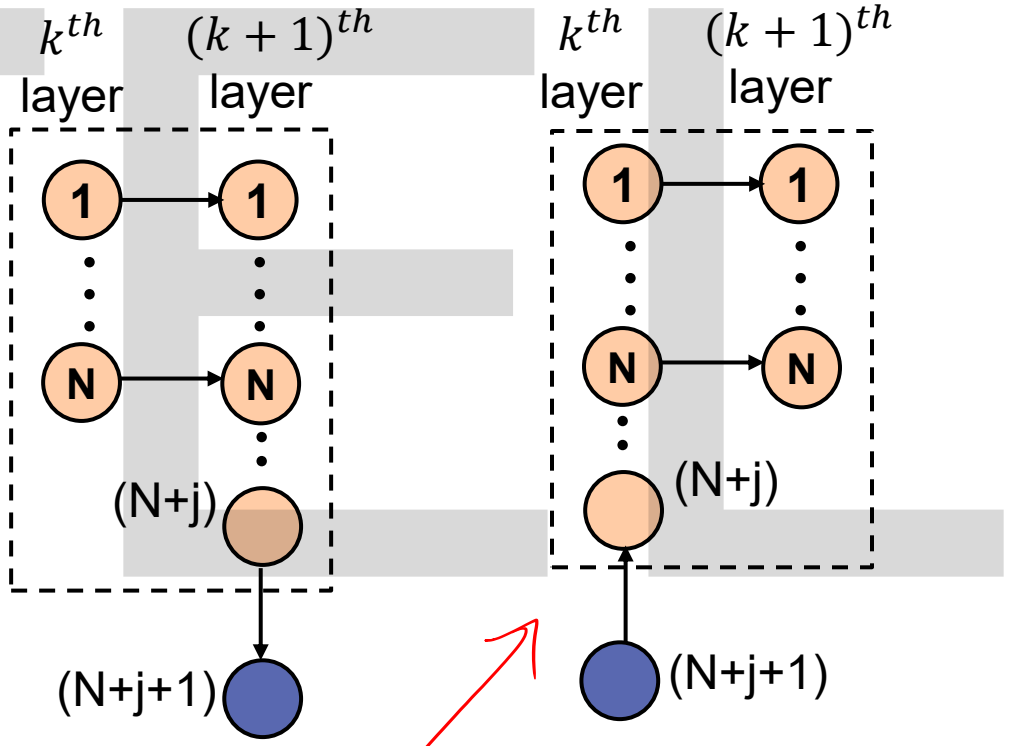
**Downward Vertical Link:** $L_{k+1,n}$-$L_{k+1,n+1}$ in $m^{th}$ cycle carries the transaction

$T_{k,n-m+2}$-$T_{k+1,n+1}$, if the transaction exists



# Other Cases

- **Also constructed the schedules when the number of the routers in two consecutive layers are not equal**
- **Detailed proof is shown in [1]**
- **We obtain minimum possible communication latency for a give DNN**




[1] Mandal, Sumit K., et al. "A latency-optimized reconfigurable NoC for in-memory acceleration of DNNs." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 10.3 (2020): 362-375.

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# Reconfigurable NoC (1)

- **There are two class of CNNs**
  - One is used in edge computing hardware
  - Another is used in cloud-based computing platform
- **We observe similarity in optimal NoC for different networks in each of the two class**
  - Similarity in number of routers for each layer (shown in next slide)
- **On the chip, a number of routers will be reserved for each layer**
- **Before executing the NN algorithm, the proposed algorithm is run (on the chip) to reconfigure the NoC**
  - Deciding the number of routers needed for each layer
  - Deciding the optimal schedule

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# Reconfigurable NoC (2)

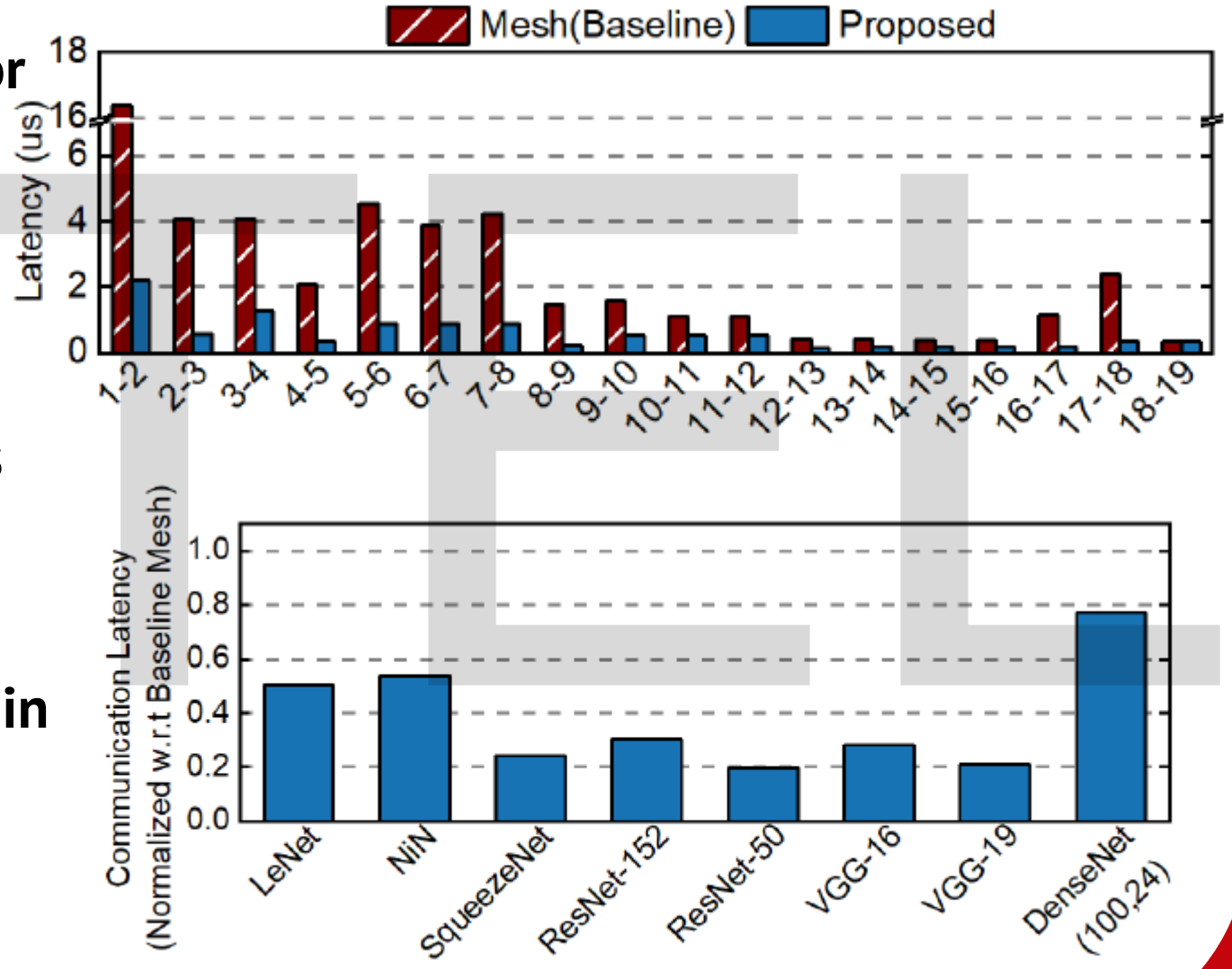
| | lenet-5 | nin-12 | squeeze-26 | vgg-16 | vgg-19 | generic_noc |
|---|---|---|---|---|---|---|
| | | | | | | |
| Layer 1 | 4 | 2 | 5 | 4 | 6 | 6 |
| 2 | 4 | 2 | 5 | 4 | 6 | 6 |
| 3 | 4 | 2 | 4 | 4 | 5 | 5 |
| 4 | 2 | 2 | 6 | 4 | 5 | 6 |
| 5 | | 2 | 6 | 4 | 4 | 6 |
| 6 | | 2 | 4 | 4 | 4 | 4 |
| 7 | | 2 | 4 | 4 | 4 | 4 |
| 8 | | 2 | 4 | 4 | 4 | 4 |
| 9 | | 2 | 3 | 4 | 3 | 4 |
| 10 | | 2 | 4 | 4 | 3 | 4 |
| 11 | | | 4 | 4 | 3 | 4 |
| 12 | | | 3 | 4 | 3 | 4 |
| 13 | | | 3 | 4 | 2 | 4 |
| 14 | | | 3 | 2 | 2 | 3 |
| 15 | | | 2 | 2 | 2 | 2 |
| 16 | | | 3 | 3 | 2 | 3 |
| 17 | | | 3 | | 2 | 3 |
| 18 | | | 2 | | 2 | 2 |
| 19 | | | 3 | | 2 | 3 |
| 20 | | | 3 | | | 3 |
| 21 | | | 2 | | | 2 |
| 22 | | | 3 | | | 3 |
| 23 | | | 3 | | | 3 |
| 24 | | | 2 | | | 2 |
| 25 | | | 3 | | | 3 |
| 26 | | | 3 | | | 3 |

- **Entries in the table shows optimal number of routers required for different networks (proposed approach)**
- **The right-most column (generic_noc) shows number of routers allocated per layer on the chip**
  - Maximum of all networks for that layer
  - The chip can afford all the 5 CNNs shown in the table
- **At runtime, for a given CNN, number of routers required for each layers of the CNN and the corresponding schedules is computed with the proposed technique**
  - Reconfigurability
- **A given network can use only the maximum available routers for the corresponding layer**
  - Can lead to sub-optimality
  - Should still be optimal than mesh-based solution
- **This hardware can support CNNs with up to 26 layers**
- **Similar common features for cloud-computing based NNs**

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# Experimental Results [1]

- **Layer-wise improvements shown for VGG-19**
  - **Up to 8× improvement in communication latency**
- **Improvement in communication latency is shown for different DNNs**
  - **Normalized with respect to the baseline communication latency**
- **Results in up to 25% improvement in inference latency**




[1] Mandal, Sumit K., et al. "A latency-optimized reconfigurable NoC for in-memory acceleration of DNNs." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 10.3 (2020): 362-375.




# THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)